

Foundations of Applied Machine Learning

ECE 4608 / 5608 — Course Introduction

Mathematical foundations, core algorithms, and evaluation practices of classical machine learning —
with an introduction to neural networks and deep learning.

Dr. Jonathan Jaramillo · Vanderbilt University

Who Should Take This Course

Curious builders

You want to actually **implement** machine learning models, not just hear about them — weekly hands-on coding is core to the course.



Comfortable with the basics

No ML background assumed — but you should be ready to work with **linear algebra, probability, and calculus**. We review all three early in the course.



Ready to talk, not just listen

This is a **flipped classroom**. Class time is discussion and problem-solving, not lecture — you need to show up prepared to engage.

What This Course Is For

Practical implementation skills

Weekly coding assignments build the muscle memory for real ML work — data cleaning, model fitting, and evaluation in Python and scikit-learn.

Assignments are completion-graded, not correctness-graded — they're for learning, not scoring.

Conceptual mastery

Weekly quizzes, structured in-class discussion, and two group projects with an oral defense confirm that **you** understand the material — not just a tool acting on your behalf.

By the end: you can build, evaluate, and **explain** a classical ML pipeline end to end.

Course Format: A Flipped Classroom

Lectures are pre-recorded. Class time is discussion, problem-solving, and collaborative coding.

Before Mon / Wed

Watch the pre-recorded lecture and complete a short, low-stakes comprehension check.

Monday / Wednesday

Discussion, problem-solving, and group work. Cold-calling and group report-outs — not optional review time.

Friday

A 10–15 minute quiz (this week's material, plus cumulative questions), then in-class coding lab time.

What's In the Course — Foundations to Evaluation

Topics build in this order — exact pacing may shift as the semester goes.

- 1 Mathematical Foundations** — linear algebra, probability & statistics, calculus for optimization
- 2 Core ML Concepts** — learning types, bias-variance tradeoff, train/val/test splits, overfitting, loss functions
- 3 Supervised Learning: Regression** — linear & polynomial regression, ridge/lasso regularization
- 4 Supervised Learning: Classification** — logistic regression, k-NN, Naive Bayes, decision trees, SVMs
- 5 Model Evaluation** — accuracy, precision/recall/F1, ROC-AUC, confusion matrices, common pitfalls · *midterm project follows*

What's In the Course — Ensembles to Neural Networks

Topics build in this order — exact pacing may shift as the semester goes.

6 Ensemble Methods — bagging & random forests, boosting, why ensembles work

7 Unsupervised Learning — k-means & hierarchical clustering, PCA / t-SNE / UMAP, Gaussian mixture models

8 Neural Networks & Deep Learning Basics — perceptrons, backprop, activations, a light touch on CNNs/RNNs

What's In the Course — Optimization to Capstone

Topics build in this order — exact pacing may shift as the semester goes.

9 **Optimization in Practice** — SGD, momentum, Adam, hyperparameter tuning strategies

10 **Broader Context** — fairness, bias, interpretability, ethics, classical ML vs. modern deep learning

11 **Capstone / Project Time** — full end-to-end final project: data cleaning, model selection, evaluation, write-up · *final project follows*

How You're Graded

Quizzes

25%

Participation

20%

Pre-lecture checks / checkpoints

10%

Midterm project + defense

15%

Final project + defense

30%

Weekly coding assignments are **not graded for correctness** — quizzes and project defenses are what confirm the understanding is yours.

Projects: Pairs, With an Oral Defense

Midterm — 15%

After Module 5 (Model Evaluation). Scoped to regression, classification, and evaluation metrics.

Both partners are questioned on **both halves** of the project — not just their own contribution.

Final — 30%

The capstone deliverable, drawing on the full course — ensembles, unsupervised learning, and neural network basics included.

Pairs may be reshuffled or continued from the midterm. Emphasis on design choices and explaining the work live.

A brief private peer evaluation is submitted after each project to help ensure fair credit within pairs.

AI Policy

⚠ **The premise:** AI coding assistants can complete most weekly assignments end to end. Grading is built around that reality, not in denial of it.

✓ AI tools allowed

Weekly coding assignments and project **implementation** (debugging, boilerplate, library exploration).

✗ No AI tools

Quizzes (closed-book, in class) and pre-lecture comprehension checks.

Project design decisions, analysis, and interpretation must be your own. If you can't explain a design choice or a piece of code in the oral defense, it counts against your grade — regardless of who or what wrote it.

Logistics

📖 **Textbook** — *Hands-On Machine Learning with Scikit-Learn and PyTorch*, Aurélien Géron (O'Reilly, 2025)

💻 **Hardware** — a laptop is required every session (tablets aren't sufficient)

</> **Software** — Python 3 and scikit-learn

🔗 **Recommended** — a paid/Pro AI assistant subscription (Claude or ChatGPT), given the AI Policy

Late work: **10% off per day** without prior instructor permission · Two unexcused absences before participation is impacted

Meet Your Instructor



BS, Physics & Computer Science — Houghton University, 2017



Systems Engineer, Lockheed Martin — signal processing & sensor fusion for aerospace platforms



PhD, Electrical & Computer Engineering — Cornell University; robotics, computer vision, and AI, centered on digital agriculture



Faculty, Cornell Systems Engineering — taught Robotics, IoT, Embedded Systems, and Cyber-Physical Systems; led multiple open-source robotics platforms; partnered with SparkFun Electronics, Experiential Robotics, Arm, and FIRST Robotics

Welcome to the Course

Let's build something real together

Dr. Jonathan Jaramillo · Foundations of Applied Machine Learning

Questions welcome — see office hours and contact info on the syllabus